Impact of Characterization on Cross-Calibration Performance for Multispectral Sensors With SI-Traceable Satellite Mission TRUTHS

Madeline Stedman, Samuel E. Hunt, Pieter De Vis, Richard Bantges, Helen Brindley, and Nigel Fox

***Abstract*—A new generation of satellites designed for low-uncertainty, SI-traceable measurements—termed "SITSats"—marks a major advancement in Earth observation (EO) capability. These missions aim to enhance the performance and interoperability of the EO "system of systems." Among them, the ESA Earth Watch Traceable Radiometry Underpinning Terrestrial- and Helio-Studies (TRUTHS) mission is designed to serve as a "gold-standard" radiometric reference for cross-calibrating EO sensors in the solar reflective domain. In this work, uncertainties in cross-calibration comparisons arising from sensor characterization and design are investigated. A processing chain to prepare collocated data for uncertainty-quantified comparison is presented. This includes steps to perform spectral band adjustment and spatial resampling. Using the TRUTHS hyperspectral imaging spectrometer (HIS) as the reference and Sentinel-2 multispectral imager (MSI) as the target, a simulation study based on high-resolution imagery assesses achievable comparison performance. A subset of uncertainty effects driven by sensor characterization is propagated through the spectral and spatial processing using a Monte Carlo approach. Sentinel-2 data are assumed at 10-m resolution, which is most sensitive to the errors considered. The results highlight the importance of sensor characterization, particularly inherent in-flight wavelength knowledge for target sensors, in such comparisons. Results from the simulation analysis give uncertainty estimates ($k = 1$) of 0.31% (blue), 0.50% (green), and 0.23% (red) for the combined error effects arising from sensor characterization and geolocation uncertainty for comparisons over the Libya-4 desert pseudo-invariant calibration sites (PICS) using an instantaneous 205-m square comparison region. Results for more heterogeneous scenes, such as rainforest, still achieve uncertainties of 0.6%–1.2% for the red–green–blue (RGB) bands over a 200 × 200 m area. The uncertainty is driven largely by the spectral component—up to 1% due to the inherent Sentinel-2 wavelength knowledge of 1 nm across various representative scenes outside of the atmospheric absorption bands. While the impact of these uncertainties may decrease when considering a diverse range of scene types, they introduce systematic errors when scenes share similar spectral characteristics. The impact of some uncertainty contributions, for example, geolocation uncertainty, is shown to be substantially reduced by aggregating samples over larger regions or over longer time periods. This analysis supports the development of low-uncertainty, ideally SITSat-enabled intercalibration approaches needed to ensure radiometric consistency across missions for generating long-term climate data records.**

***Index Terms*—Calibration, climate, cross-calibration, Earth observation (EO), hyperspectral, intercalibration, interoperability, multispectral imager (MSI), sensor characterization, Sentinel-2, SI-traceable, SI-traceable satellite (SITSat), Traceable Radiometry Underpinning Terrestrial- and Helio-Studies (TRUTHS), uncertainty.**

I. Introduction

REMOTE sensing plays a crucial role in monitoring and understanding the Earth's climate, providing essential data for policy-making, climate modeling, and climate change mitigation strategies [1]. The current Earth observing system primarily consists of small evolutions of long-established space agency satellites, which offer high-quality decadal time series, together with an increasing number of "low-cost" commercial satellites that deliver higher-resolution data both spatially and temporally as part of multisatellite constellations. Ensuring the interoperability of this observing system requires, as a minimum, harmonized and consistent data, meaning that any biases observed between sensor-measured radiances and reflectances are understood and correctable. The largest source of such biases stems from differences in the underpinning calibration of the sensors' radiometric performance, or knowledge of, in-flight. This is of particular importance when considering the demanding requirements of climate observation [2]—which necessitates consistent radiometric measurements from a series of satellites to create sufficiently long time-series, with the comprehensive temporal and spatial sampling needed to enable very small signals/trends to be detectable as needed for reliable climate monitoring. In response to these demands, advancements in Earth observation (EO) capabilities over the last decade include the development of an upcoming class of missions with high-accuracy onboard SI-traceable calibration systems embedded into the design. This new generation of

Received 26 July 2025; revised 28 October 2025; accepted 13 November 2025. Date of publication 17 November 2025; date of current version 4 December 2025. This work was supported in part by the Traceable Radiometry Underpinning Terrestrial–and Helio–Studies (TRUTHS) Mission Accompanying Consolidation (TMAC) and TRUTHS mission Accompanying Consolidation Toward Operations Study (TACOS) Contracts through European Space Agency, European Space Research Institute (ESA-ESRIN) under Grant 4000146584/25/NL/J; in part by the Natural Environment Research Council (NERC "U.K. EO Climate Information Service") under Grant NE/X019071/1; and in part by the National Measurement System Programme of U.K. Government's Department for Science, Innovation and Technology. *(Corresponding author: Madeline Stedman.)*

Madeline Stedman is with the National Physical Laboratory, Teddington, TW11 0LW Middlesex, U.K., and also with the Imperial College London, SW7 2AZ London, U.K. (e-mail: maddie.stedman@npl.co.uk).

Samuel E. Hunt, Pieter De Vis, and Nigel Fox are with the National Physical Laboratory, Teddington, TW11 0LW Middlesex, U.K.

Richard Bantges and Helen Brindley are with the Imperial College London, SW7 2AZ London, U.K.

Digital Object Identifier 10.1109/TGRS.2025.3633954

© 2025 The Authors. This work is licensed under a Creative Commons Attribution 4.0 License.
For more information, see https://creativecommons.org/licenses/by/4.0/

satellites has been termed "SI-Traceable Satellite" (SITSat) by the Committee on EO Satellites (CEOS) and the World Meteorological Organization's Global Space Inter-Calibration System (WMO-GSICS), accompanied by a formal definition [3].

SITSats currently in development include ESA Earth Watch satellite mission: traceable radiometry underpinning terrestrial and Helio-studies (TRUTHS) [4], NASA's climate absolute radiance and refractivity observatory (CLARREO) Pathfinder [5]—designed for deployment on the international space station (ISS) in the next few years—and the China Meteorological Administration (CMA)'s satellite mission LIBRA [6]. These missions are being developed with a primary objective of providing a space-borne "gold-standard" reference against which to calibrate other EO satellite sensors in the solar-reflective domain. By disseminating their low-uncertainty, SI-traceable calibration, these missions aim to enhance the performance of sensors operating in this domain and support the generation of harmonized, consistent datasets required for accurate climate monitoring.

TRUTHS has been developed to date by ESA on behalf of a consortium of European nations led by the U.K. [4]. Key objectives of the mission are to provide benchmark measurements of reflected solar radiation at the top-of-atmosphere (TOA) to monitor change and to serve as a calibration reference for other sensors. TRUTHS will achieve this using its hyperspectral imaging spectrometer (HIS), delivering spectrally resolved measurements across 320–2400 nm with a target uncertainty of 0.3% ($k = 2$). The instrument is primarily nadir-viewing, with the potential for some off-nadir observations. TRUTHS will achieve traceability to the SI in orbit through its innovative onboard calibration system (OBCS), which includes, for the first time, an onboard primary standard, the cryogenic solar absolute radiometer (CSAR). This system effectively mirrors laboratory calibration methods used by National Metrology Institutes (NMIs) on Earth [7]. In addition to Earth-viewing measurements, TRUTHS will measure total solar irradiance (TSI), spectral solar irradiance (SSI), and spectral lunar irradiance (SLI). The mission's design, including its polar, non-Sun-synchronous, precessing orbit and hyperspectral capabilities (2–8-nm spectral resolution), is optimized in part to support cross-calibration. The orbit enables diurnal sampling of the planet for climate monitoring and results in a diverse set of matchups—near simultaneous overpasses with other satellites used as cross-calibration targets—across a wide range of geographical and illumination conditions. Additionally, the hyperspectral resolution of TRUTHS allows for more accurate modeling of the spectral bands of other missions through improved convolution with their spectral response functions (SRFs).

The primary technique that will be used to cross-calibrate EO sensors with TRUTHS in-flight is comparison of measurements within satellite-to-satellite "matchups." A matchup is defined as an event where two satellite sensors observe the same location at the same time, within a given tolerance. In the ideal case where both sensors are viewing close to nadir, this is commonly referred to as a simultaneous-nadir overpass (SNO).

Analysis of matchups is a widely used method for comparing and/or intercalibrating satellite sensors in-flight. This approach has been extensively applied to compare multispectral sensors, as demonstrated in numerous studies, such as [8] and [9], and more recently to cross-calibrate sensors against hyperspectral reference sensors [10].

Currently, uncertainty budgets for cross-calibration are primarily dominated by the measurement uncertainty of the reference sensor [11], [12]. Furthermore, the degree of confidence in the resultant uncertainties and their traceability to SI are not always sufficient to meet the demanding requirements necessary for accurate climate monitoring [3].

SITSats can, in principle, drive significant advancements, as, with the improved level of uncertainty they aim to achieve, cross-calibration performance will no longer be limited by the uncertainty of the reference measurements. Instead, the limiting factor will shift to the challenges associated with the differences between observations of the two sensors during matchups and limitations resulting from the design and characterization of the sensor under test. In the case of optical observations this can include effects related to the observational configuration [13], such as viewing geometry mismatch (resulting in changes in surface reflectance and polarization) [14], [15], and temporal mismatch (resulting in changes in illumination angle and intermediate atmospheric path) [16], [17], and effects intrinsic to the pair of sensors and comparison process itself, such as mismatch in SRFs [18], [19] [20] and spatial sampling [12]. Such effects introduce a comparison mismatch which adds further uncertainty to the comparison beyond the inherent uncertainties of the reference sensor's radiometric performance, resulting from its own calibration. Therefore, several adjustments and/or corrections must be applied to compensate for this comparison mismatch and make the sensor data more comparable.

Previous studies have emphasized that rigorous sensor characterization and calibration are critical to minimize uncertainties introduced by the cross-calibration process itself [12]. In particular, spectral uncertainties are shown to be of particular importance to the overall uncertainty associated with any spectral adjustment performed to account for differences in the SRFs of the sensors. This is because they can introduce systematic errors when the spectral characteristics of the observed scene differ, even slightly, from those of the calibration reference. While most existing studies focus on comparisons between multispectral sensors using spectral band adjustment factors (SBAFs), TRUTHS HIS, as a hyperspectral instrument, offers the potential for improved reconstruction of target sensor bands, reducing spectral adjustment uncertainties.

Previous work has shown that geometric and temporal mismatches between TRUTHS and Sentinel-2 observations can be mitigated to below 0.1% over most land areas using realistic thresholds [13], although some scene types are more sensitive to these effects. This study builds on that by focusing on ideal matchup conditions—simultaneous, nadir-viewing observations—to isolate and assess uncertainties arising from sensor characterization and design, such as wavelength knowledge and geolocation uncertainties. These uncertainties are propagated through a metrology-based framework to estimate their impact on the final comparison using a set of representative simulated, simultaneous nadir matchups to esti-

mate a lower bound on the achievable uncertainty. A future refinement of this study will build on previous work to more fully account for the effects of geometric and temporal mismatches.

The simulation methodology is outlined in Section II-A, with the target sensor data prepared for comparison described in Section II-B. The processing algorithm to prepare TRUTHS and target sensor data for comparison is described in Section II-C, and the uncertainty propagation approach is covered in Section II-D. Individual uncertainty contributions from each processing step are evaluated separately and then combined in quadrature to estimate the total uncertainty from sensor-characterization-driven effects. Results of the simulation study are presented in Section III.

II. Methodology

Preparing satellite matchup data for comparison requires several processing steps to account for differences in how each sensor observes the scene. As introduced in Section I, for optical sensors, this involves, at a minimum, aligning the observations both spectrally and spatially to address differences in sampling. The methodology presented here aims to evaluate the uncertainty introduced by these steps through a set of simulated matchup scenarios. TRUTHS is used as the reference sensor, with the Sentinel-2 multispectral imager (MSI) as the target. MSI is chosen for its relevance as an anticipated user of TRUTHS' sensor-to-sensor calibration system and its representativeness of similar multispectral sensors.

Section II-C introduces the sensor-to-sensor comparison processor (S2SC), a processing algorithm currently under development for TRUTHS. The S2SC is designed to compensate observational differences between sensor pairs by generating spatially and spectrally aligned "comparison samples"—pixel measurements from each image that have been individually aggregated such that the two sensors are observing a matching field of view (FOV) and spectrally convolved to be similar. This framework provides a consistent basis for evaluating uncertainties arising from sensor characterization, such as wavelength knowledge and geolocation accuracy.

The overall performance of the cross-calibration is inherently dependent on the generation and subsequent evaluation of the comparison samples. The processing performed to enable comparison of these samples introduces additional uncertainties beyond the inherent calibration uncertainties of the reference sensor. It is, therefore, essential to evaluate the uncertainty introduced at each step to accurately assess the total uncertainty associated with the comparison.

The following notation is used throughout.

1) Variables representing radiance are denoted as L, with subscripts indicating the sensor (e.g., HIS and MSI) and other relevant parameters.
2) Spectral quantities are indicated with a bar (e.g., spectral radiance is represented as $\underline{L}$.
3) Raster imagery is denoted with a hat (e.g., raster radiance is represented as $\hat{L}$).

A. Simulation Analysis

Fig. 1 provides a high-level overview of the approach used to test the performance of the TRUTHS S2SC algorithm using simulated data that represent the TRUTHS HIS and Sentinel-2 MSI comparisons.

As illustrated in this figure, the simulation analysis comprises two main steps.

1) *Simulated Observation Preparation:* Simulated (mock), collocated observations of the reference and target sensors are prepared based on a set of representative matchup scenarios. Each simulation involves generating a reference Earth scene based on reference data (i.e., satellite imagery, using a radiative transfer model) and observing it through forward models of the two sensors (see Section II-B). Scenes are designed to represent different surface types, as uncertainty contributions typically vary with scene characteristics.
2) *Algorithm Testing:* The S2SC, introduced in Section II-C, is applied to the simulated matchups to evaluate comparison samples. Input uncertainties related to instrument characterization and design are propagated through the algorithm using the Monte Carlo method (MCM), following GUM Supplement 1 [21]. The uncertainty propagation approach is detailed in Section II-D.

Uncertainty contributions from both the spectral and spatial processing steps are assessed, considering error correlations along both dimensions. To achieve this, the simulated scenes must capture spectral and spatial information at a finer resolution than the sensors under test. Separate simulation cases address spectral and spatial effects using different sources of reference data; one set is based on high spectral resolution data, and the other on high spatial resolution data. Since spectral and spatial errors are expected to be uncorrelated, their uncertainties can be propagated independently and combined in quadrature to give the total uncertainty for the effects considered. The formulation of the simulated scenes used in this study is detailed in Section II-B.

B. Observation Simulation

The TOA radiance imagery simulated for this analysis is designed to replicate real-world matchup scenarios between the TRUTHS HIS and Sentinel-2 MSI sensors across various Earth scenes, enabling a detailed analysis of the uncertainties involved in cross-calibration. The Earth scene modeling approach makes use of high-resolution data from a variety of sources combined with a radiative transfer model. This is combined with a simple instrument model for each of the sensors, described in Section II-B2, to generate representative matchup datasets for use in subsequent performance analysis.

1) Scene Model: The simulated Earth scenes generated are detailed in Table I. The formulation of these scenes is described in the following sections, first for the spectral scenes in Section II-B2a and then for the spatial scenes in Section II-B2b.

a) Spectral Scenes: A set of simulated TOA radiance scenes was generated using high-spectral-resolution data (see

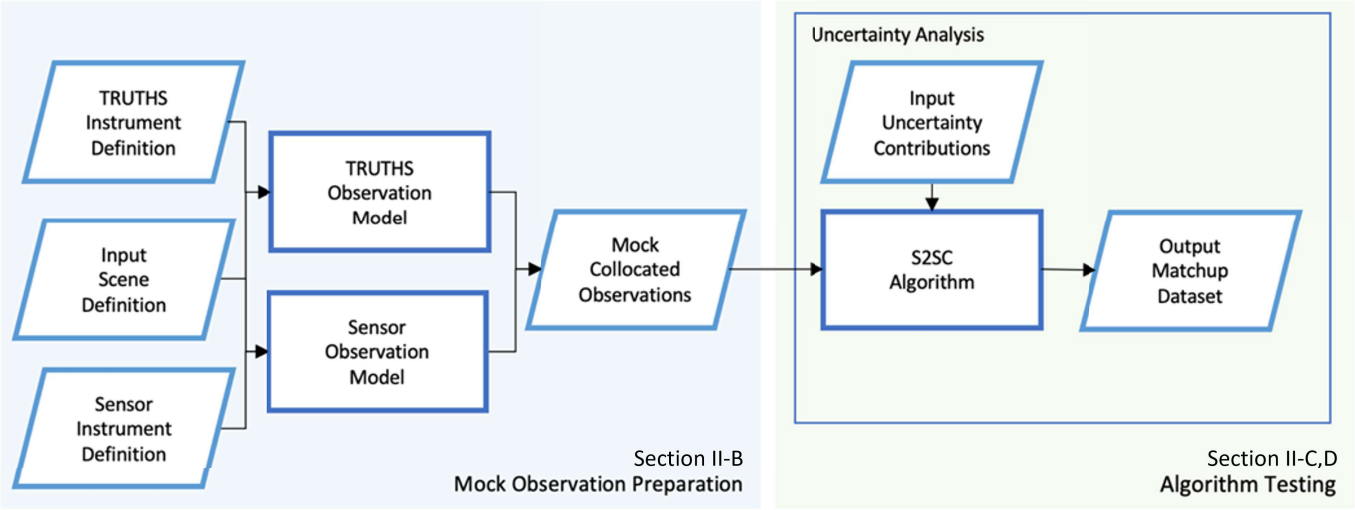


Fig. 1. Simulation model for the S2SC performance assessment of cross-calibration performance. The diagram is divided into two main steps: *mock observation preparation* (blue, explained in Section II-B) and *algorithm testing* (green, S2SC algorithm explained in Section II-C); uncertainty analysis explained in Section II-D.

TABLE I
SIMULATIONS PERFORMED TO ASSESS PERFORMANCE OF SPECTRAL AND SPATIAL PROCESSING STEPS—REFERRED TO AS SPECTRAL SCENE SIMULATIONS AND SPATIAL SCENE SIMULATIONS

Step	Instrument Characterisation Effects Under Test	Simulated Scene Types	Sources of Reference Data
Spectral	Reference and target sensor wavelength knowledge and instrument noise	**Desert:** Based on Libya-4 PICS, a widely used calibration site [22]–[24]. **Rainforest:** Based on the Amazon. **Antarctic Snow:** Based on Dome-C PICS, another established calibration site [14]. **Deep Convective Cloud (DCC):** A common calibration target in the visible and near-infrared spectral domains [25].	**For Desert, Rainforest, Snow:** Geolocated surface reflectance PRISMA L2A/D data, spectrally interpolated and propagated to TOA using the `libRadtran` radiative transfer model with realistic atmospheric parameters (including total column water vapour, aerosol optical depth, total column ozone, atmospheric pressure) and observation geometry. **For DCC:** Sentinel-2 MSI L1C TOA radiance data (identified as DCC in [26]) and high-resolution SCIAMACHY radiance spectrum averaged over DCC pixels [27].
Spatial	Reference and target sensor geolocation uncertainty, and instrument noise	**Desert:** Libyan desert region centred around Libya-4 PICS. **Rainforest:** Based on the Amazon. **Antarctic Snow:** Based on Dome-C PICS.	Geolocated orthorectified Airbus Pléiades L1A TOA radiance.

Table I). These are used to test the performance of the S2SC spectral adjustment algorithm. The scene types were chosen to be representative of typical sites used for cross-calibration. A range of representative scenes is simulated, capturing a range of surface characteristics, as the dominant effects differ between scenes [14], [23]. In particular, the inclusion of vegetated scenes enables exploration of the sensor's linearity, as well as sensitivity to relative spectral intensities.

For desert, rainforest, and Antarctic snow, Earth scenes are based on cloud-free hyperspectral PRISMA L2D surface reflectance data [28] propagated to TOA using the radiative transfer model `libRadtran` [29]. Radiative transfer in `libRadtran` is run with a bandwidth of 1cm^{-1} (corresponding to 0.01–0.6 nm for the HIS wavelength range), interpolating the PRISMA spectral bands. This ensures that atmospheric molecular absorption features are sufficiently oversampled spectrally with respect to HIS sampling, while PRISMA's ~10-nm resolution is considered sufficient to capture surface reflectance spectra, which are assumed to vary smoothly with wavelength. Both the visible and near-infrared (VNIR) and short-wave infrared (SWIR) bands of PRISMA are used to cover as much of TRUTHS HIS' anticipated spectral range (320–2400 nm) as possible. Although PRISMA's spectral coverage begins at 400 nm—omitting the lower portion of

TABLE II

ATMOSPHERIC PARAMETERS USED IN RADIATIVE TRANSFER SIMULATIONS FOR TOP OF ATMOSPHERE (TOA) PROPAGATION FOR EACH SCENE TYPE. VALUES ARE DERIVED FROM A COMBINATION OF LITERATURE [30], [31], [32] AND NEARBY AERONET INSTRUMENTATION [33], WHERE AVAILABLE. WHERE SPECIFIC PARAMETERS ARE NOT LISTED IN THE TABLE, DEFAULT VALUES FROM LIBRADTRAN ARE USED

Parameter	Desert	Rainforest	Snow	Notes / Units
Atmospheric profile	US-standard	Tropics	Subarctic Winter	Standard atmospheres in `libRadtran` [29]
Surface pressure	962	1017	645	hPa, at 0 km altitude
Total column water vapour (TCWV)	15	50	0.5	mm
Total column ozone (TCO)	277.5	277.5	110	DU
Aerosol model/type below 2 km	Continental Avg.	Continental Avg.	Antarctic	Aerosol mixtures in `libRadtran` [29]
Aerosol model/type above 2 km	Background	Background	Background	Aerosol mixtures in `libRadtran` [29]
AOD at 550 nm	0.1	0.2	0.08	Dimensionless

HIS' range—this is not a limitation for the current study, as Sentinel-2 MSI has no bands below 400 nm. In the propagation of the PRISMA surface reflectance to TOA, realistic atmospheric parameters for each scene were taken from a combination of literature [30], [31], [32] and nearby AERONET instrumentation [33] where available—values used are given in Table II.

For DCC scenes, a different simulation approach was used due to the lack of available PRISMA data. It was instead based on multispectral TOA reflectance from Sentinel-2 MSI, identified in literature as a DCC [26], combined with a high spectral resolution reference spectrum derived from averaged SCIAMACHY observations [27].

The Earth scene spectral simulations generate TOA radiance images that are oversampled spectrally with respect to TRUTHS, at 0.01-nm resolution, and spatially interpolated from the 30-m resolution of PRISMA surface reflectance to 1-m resolution using linear interpolation. This interpolation assumes that no significant spatial detail exists between the 30-m input resolution and the 1-m output scale. As such, the simulations are not intended to reproduce fine spatial structure, but rather to support the evaluation of spectral effects; they will ultimately be rescaled to the coarser spatial resolutions of the target sensors.

b) Spatial Scenes: To assess the impact of effects affecting the spatial resampling, collocated TOA radiance imagery representative of TRUTHS HIS and Sentinel-2 MSI in the spatial domain is simulated. As before, this is done for Libyan desert, rainforest, and Antarctic snow scenes (see Table I).

The simulations are based on Airbus Pléiades L1A TOA radiance data (CNES (2024), Distribution Airbus DS). With a native resolution of 2 m, this data provides the underlying high-resolution spatial information for the scene, oversampled with respect to the resolutions of HIS (50 m) and MSI (10 m). It covers a wavelength range of 450–915 nm across four multispectral bands.

2) Instrument Model: To simulate the measurements of TRUTHS HIS and Sentinel-2 MSI, a simplified instrument model is applied to the simulated Earth scenes output by the Scene Model.

MSI, the target sensor considered for this analysis, is a push-broom multispectral spectrometer onboard the ESA Sentinel-2 satellite series [34]. It measures the Earth's reflected radiation across the VNIR and SWIR spectral domains using 13 spectral bands. The spatial resolution varies by band.

1) *60 m:* Bands 1, 9, and 10—primarily used for atmospheric corrections and cloud screening, including aerosol retrieval (443 nm), water vapor correction (940 nm), and cirrus detection (1375 nm).
2) *10 m:* Bands 2, 3, 4, and 8—corresponding to the blue, green, red, and near-infrared (NIR) bands. These bands support a range of applications, including the monitoring of vegetation, soil, and water cover.
3) *20 m:* Bands 5, 6, 7, 8A, 11, and 12—focused on vegetation red-edge and SWIR observations.

For simplicity, in this analysis, it is assumed that all MSI bands have a spatial resolution of 10 m, and the HIS bands have a resolution of 50 m (the nominal resolution proposed for use over designated cal/val sites). The finest resolution is used for all MSI bands, as this is expected to be the most sensitive to spatial effects considered in this analysis. TRUTHS HIS is, therefore, considered the lower-resolution sensor in this example.

For the spatial scenes, it is assumed that the spectral characteristics of the Airbus Pléiades multispectral bands are sufficiently representative of sensors like MSI, in terms of capturing the relevant spatial heterogeneity. As such, only spatial resampling is applied to prepare representative inputs for the sensitivity analysis. The high-resolution Pléiades data (2 m) is downscaled via area averaging to approximate resolutions of 10 and 50 m, corresponding to those of MSI and HIS, respectively—analogous to the approach used in the spectral scene simulations. Since exact downscaling to 10 and 50 m is not always possible due to the 2 m Pléiades resolution, the nearest achievable resolution is used instead.

C. Truths Sensor-to-Sensor Comparison Algorithm

In Fig. 2, the processing chain to prepare the TRUTHS HIS and Sentinel-2 MSI collocated data for comparison is illustrated.

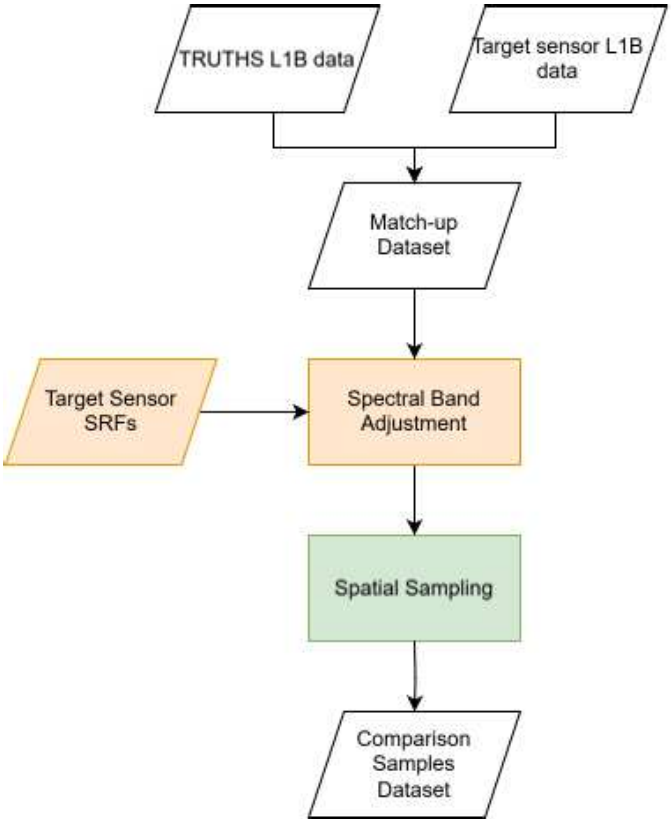


Fig. 2. Matchup processing chain to prepare collocated observations from TRUTHS HIS and a target sensor for comparison.

The sensor measurement data is processed to make them more comparable through the following steps.

1) Integrating the hyperspectral TRUTHS HIS imagery to match the multispectral bands of the sensor under test, $\underline{\lambda}_{\text{target}}$, to give $\underline{\hat{L}}_{\text{HIS}}(\underline{\lambda}_{\text{target}})$.
2) Sample the data spatially by combining individual pixel measurements such that the two sensors observe the same FOV, to give sensor and reference measurements, $\underline{L}_{\text{target},m}(\underline{\lambda}_{\text{target}})$ and $\underline{L}_{\text{HIS},m}(\underline{\lambda}_{\text{target}})$, for each comparison sample, m.

Sections III-C1 and III-C2 define the algorithms used to perform each of these.

1) Spectral Band Adjustment: The first step is to adjust the spectral bands of the TRUTHS HIS image within the collocated region, $\underline{\hat{L}}_{\text{HIS}}(\underline{\lambda}_{\text{HIS}})$, so they match those of the sensor under test to give $\underline{\hat{L}}_{\text{HIS}}(\underline{\lambda}_{\text{target}})$. Each spectral band of the target sensor is defined by its SRF, $\text{SRF}_i(\underline{\lambda}_{\text{SRF},i})$, where i represents the band index. In general, the appropriate method for spectral adjustment depends on the relative spectral resolutions of the sensors being compared. The method described below applies unless the sensor resolution is similar to or finer than that of TRUTHS. Alternative methods can be used for these special cases, but they will be addressed in a subsequent paper.

For the HIS-MSI matchups considered in this article, the hyperspectral TRUTHS HIS spectrum is integrated into the target sensor (MSI) spectral bands by applying the following algorithm for each band.

a) Interpolate the Truths HIS Radiance Spectrum: First, the HIS radiance spectrum must be interpolated to the wavelength grid of the target sensor SRF, that is, to evaluate $\underline{L}_{\text{HIS}}(\underline{\lambda}_{\text{SRF}_{\text{target},i}})$, to enable the numerical integration in the following step. The SRF wavelength grid is defined according to the spectral resolution with which the SRF data are supplied for the respective mission. This is performed using cubic spline interpolation—ongoing work is assessing the performance/adequacy of this interpolation method as applied here. The spectral range is limited to the wavelengths around the nonzero region of the target sensor's SRF, extended on each side by 20% of the width of this region, to reduce computation and avoid potential numerical errors.

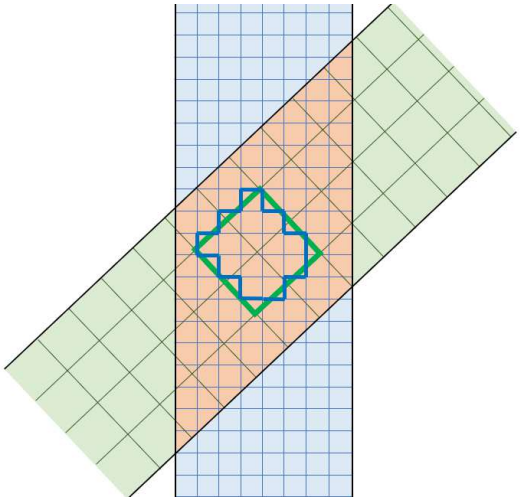


Fig. 3. Region of collocated observations (orange), for the swath of a lower resolution (green) and higher resolution (blue) satellites. The thick green line represents a field of view for a comparison sample from the lower-resolution sensor, and the thick blue line represents the higher-resolution sensor's reconstruction of this FOV.

b) Perform Numerical Spectral Band Integration: Per spatial pixel of $\underline{\hat{L}}_{\text{HIS}}(\underline{\lambda}_{\text{HIS}})$ evaluate the spectral band integral as follows:

$$L_{HIS}\left(\lambda_{target,i}\right) = \frac{\int \underline{L}_{HIS}\left(\underline{\lambda}_{SRF_{target,i}}\right) SRF_{target,i}\left(\underline{\lambda}_{SRF_{target,i}}\right) d\lambda}{\int SRF_{target,i}\left(\underline{\lambda}_{SRF_{target,i}}\right) d\lambda}. \tag{1}$$

This is performed using the trapezium rule.

The result is the TRUTHS HIS reconstruction of the sensor-under-test (MSI) spectral imagery, $\underline{\hat{L}}_{\text{HIS}}(\underline{\lambda}_{\text{target}})$.

2) Spatial Sampling: The next step performed on the matchup imagery is spatial resampling. The objective of this is to create coherent comparison samples from each sensor—pixel measurements from each image that have been aggregated such that the two sensors are observing a matching FOV. This is illustrated in Fig. 3. The set of measurements $L_{\text{HIS},m}(\lambda_{\text{target},i})$, $L_{\text{target},m}(\lambda_{\text{target},i})$ (as well as corresponding uncertainty and auxiliary data), where m is the index of each comparison sample and i is the index of each target sensor band, are established for evaluation.

This resampling is completed with the following steps for each spectral band.

a) Evaluation of Comparison Grid Resolution: The two sensors are labeled fine (F) and coarse (C), where Δx_F and Δx_C denote their respective spatial resolutions. The compar-

ison FOV is explicitly defined as an integer multiple, k_{scale}, of the coarser sensor's resolution, that is, the comparison FOV= $k_{\text{scale}}\Delta x_C$. To ensure that a sufficient number of fine-resolution (sensor F) pixels fall within each comparison FOV, a minimum count, k_{lim}, is required

$$k_{\text{lim}}\Delta x_F < k_{\text{scale}}\Delta x_C. \tag{2}$$

This is satisfied by choosing

$$k_{\text{scale}} = \left\lceil k_{\text{lim}} \frac{\Delta x_F}{\Delta x_C} \right\rceil \tag{3}$$

where the notation $\lceil\rceil$ indicates the ceiling function. The value of k_{lim} used, therefore, defines the resolution of the comparison grid. In this study, $k_{\text{lim}} = 20$ is used—shown to be an appropriate value from analysis based on high-spatial-resolution imagery. For HIS and MSI resolutions of 50 and 10 m, respectively, this corresponds to a minimum comparison resolution of $k_{\text{lim}} \times \Delta x_F = 20 \times 10 =$ an area of 200×200 m.

b) Binning of the Coarser Resolution Sensor (HIS) Data to the Comparison Grid: Based on this, the collocation image grid data of the coarser resolution sensor is binned by the factor k_{scale}. Each binned pixel forms an individual comparison sample, as

$$L_{C,m} = \frac{1}{N}\sum_{j=1}^{N} L_{C,j} \tag{4}$$

where $\{L_{C,j}\}$ are the N pixel values that contribute to a given pixel on the binned grid, where $N = k_{\text{scale}}^2$. The latitude and longitude coordinates of the comparison grid are defined as the center of each bin. Auxiliary variables are similarly binned.

c) Resampling of the Higher Resolution Sensor (MSI) Data to the Comparison Grid: Next, the higher resolution sensor is sampled onto the comparison grid, averaging the nearest-neighbor values identified using a k-dimensional tree—a spatial indexing structure commonly used for fast nearest-neighbor searches—as

$$L_{F,m} = \frac{1}{N}\sum_{n=1}^{N} L_{F,n} \tag{5}$$

where $\{L_{F,n}\}$ are the N pixel values that are closest to a given pixel on the comparison grid. As before, auxiliary variables are similarly binned.

In the case where either sensor does not completely sample the comparison FOV, for example, due to the shape of the region of collocated observations, the comparison sample is discarded.

D. Algorithm Performance Testing

The matchup comparison process introduced in Section II-C is analyzed using metrological principles based on the GUM framework [35]. Total uncertainty is estimated by evaluating all contributing error sources, with a focus here on those arising from sensor characterization. These are grouped into two categories: effects influencing spectral adjustment (Section II-D1) and those affecting spatial resampling (Section II-D2).

1) Spectral Effects: The subset of these effects impacting the spectral adjustment performed to evaluate the TRUTHS HIS reconstructions of the target sensor bands is detailed in Table III. The correlation scales and magnitudes used in the simulation analysis for the spectral adjustment are given for each of these uncertainty contributions. These values are largely taken from literature on Sentinel-2, and the TRUTHS specification (as defined at the time of writing). Although the HIS sensor noise is listed as an effect in this table, MSI noise is not included in Table III as no spectral adjustment is applied to the multispectral MSI bands in this example—MSI noise is, therefore, included in the uncertainty propagation in the following processing step, the spatial aggregation of the data.

2) Spatial Effects: Table IV lists the subset of effects impacting the spatial resampling measurement equation that are driven by sensor characterization, along with corresponding correlation forms and magnitudes used in this analysis. A combination of the satellites' geolocation knowledge uncertainties and the limitations of the resampling algorithm due to a combination of the spatial heterogeneity of the target surface and the sensors' ground sampling distance (GSD) and modulation transfer function (MTF) will contribute to the uncertainty associated with this processing. This is mitigated by binning pixels to a sufficiently large, common comparison FOV so that the pixels for both instruments can be convolved to match it with negligible impact of residual FOV differences, and any residual inhomogeneity of the surface can be minimized through averaging. This is further explored in Section III-D. As with previous values, these are primarily sourced from Sentinel-2 literature and the defined TRUTHS specification. The systematic uncertainty associated with the output of the spectral processing step is omitted here, as it is instead combined in quadrature with the systematic uncertainty from the spatial resampling to determine the total uncertainty in this analysis.

3) Uncertainty Propagation: Uncertainty propagation is performed using the MCM—numerical methods such as this offer a practical approach to propagate input quantity uncertainties through a measurement function. This is performed here using the open-source community metrology toolkit (CoMet Toolkit) [36], a Python-based framework for handling measurement uncertainties and error covariances.

There are two stages of this propagation.

1) Spectral adjustment uncertainties (Table III) are propagated through the algorithm described in Section II-C1.a, using the high-spectral-resolution simulations from Section II-C1a.
2) Spatial resampling uncertainties (Table IV) are propagated through the method in Section II-C2, using simulations described in Section II-C1.b based on high-spatial-resolution data.

These uncertainty contributions are assumed to be uncorrelated and are combined in quadrature to give the total uncertainty associated with the matchup comparison. This assumption is considered reasonable for the homogeneous scenes analyzed in this study, though the influence of potential correlations, particularly for more heterogeneous scenes, will be investigated in future work.

TABLE III
EFFECTS CONSIDERED IN THE SUBSEQUENT UNCERTAINTY ANALYSIS FOR THE SPECTRAL BAND INTEGRATION

Effect	Affected Term in Measurement Function	Correlation Form: Spectral	Correlation Form: Spatial (between comparison samples	Uncertainty Magnitude (k=1)
HIS noise	L_{HIS}	Random	Random	Wavelength-dependent*
HIS Wavelength Uncertainty**	λ_{HIS}	Systematic	Structured***	0.05 nm
MSI Wavelength Uncertainty**	λ_{MSI}	Systematic	Structured***	1 nm

*HIS signal-to-noise ratio (SNR) for albedo of 0.2 at 100 m ground sampling distance (GSD) is defined in the specification (as defined at the time of writing) to be 40 ($<$ 350 nm) and $\sim$420 (380-1950 nm). **In this study it is assumed that the spectral response function shape is well-characterised for the sensors, with the only contributing effect being the effective wavelength knowledge, as this is expected to be the dominant effect. While real-world spectrometer-based systems may exhibit some variation between bands, for this analysis a uniform spectral shift across all bands is assumed for simplicity.
***Errors arising due to wavelength uncertainty depend on the spectral shape. Therefore, in this work, they are assumed to be systematically correlated within samples of the same scene type and randomly correlated between samples of different scene types (e.g., desert vs. rainforest).

TABLE IV
EFFECTS CONSIDERED IN THE SUBSEQUENT UNCERTAINTY ANALYSIS FOR THE SPATIAL RESAMPLING PROCESSING

Effect	Affected Term in Measurement Function	Correlation Form: Spectral	Correlation Form: Spatial (between comparison samples)	Uncertainty Magnitude (k=1)
Random component of uncertainty output by the spectral adjustment processing	$L_{\text{HIS}}(\lambda_{\text{MSI}})$	Random	Random	Wavelength-dependent
MSI noise	L_{MSI}	Random	Random	Wavelength-dependent*
HIS Geolocation Uncertainty	x_{HIS}, y_{HIS}	Systematic	Systematic	25 m (1/2 pixel)
MSI Geolocation Uncertainty	x_{MSI}, y_{MSI}	Systematic	Systematic	5 m (1/2 pixel)
Surface inhomogeneity	L_{MSI}, L_{HIS}	N.A.	Structured**	N.A.***

*Sentinel-2 MSI SNR between 50 (B10) and 168 (B03) [34].
**Surface inhomogeneity varies across a scene.
***The effect of surface inhomogeneity is included in this simulation analysis through the use of real surface reflectance data to underpin the simulations.

A Gaussian PDF is assumed for all randomly correlated effects included.

III. RESULTS

This section presents the results of the simulation study. Section III-A showcases the spectral and spatial simulated images. Uncertainties are propagated through the spectral adjustment for the desert scene in Section III-B and for all scene types in Section III-C. Section III-D details the corresponding results for spatial resampling. The sensitivity of these processing steps to the target sensor (i.e., S2) wavelength knowledge and geolocation uncertainty is explored in Section III-E, assessing the impact of target sensor characterization on matchup comparison performance. Finally, the total uncertainty associated with the comparison samples due to effects driven by instrument characterization is calculated in Section III-F.

A. Simulated Imagery

Fig. 4 shows the simulated TRUTHS and MSI radiance spectra for the desert, rainforest, snow, and DCC scenes prepared as described in Section II-B, underpinned by high-spectral-resolution data. The spectra plotted are obtained by averaging the radiance over all pixels in the region of interest (ROI) for visualization purposes; however, the unaveraged images were used in the subsequent analysis. The spectra shown here have not had any further processing applied, that is, no spectral adjustment or spatial resampling—the corresponding simulated images form the input to the S2SC processing defined in Section II-C that is performed in this analysis.

Fig. 5 shows the full resolution (2 m, Pléiades resolution) TOA radiance image, and the 10- and 50-m images for a subset of the scene, as indicated, and at the 500-nm band, prepared as described in Section II-B—a subset of the imagery is shown to illustrate the resolution differences between the sensors, but the following analysis is performed using a 5×5 km ROI.

A limitation of these simulated images is that any effects present in the reference data used to underpin the simulations will propagate through to the simulated images. For example, effects such as striping and hot pixels have been identified in PRISMA surface reflectance images [37]. The extent of these spectral artifacts in the PRISMA data is described to be band-dependent. However, while visually noticeable in some of the simulated imagery, this does not have a large effect on the overall uncertainty distribution.

B. Simulation Uncertainty Analysis for Spectral Band Adjustment—Desert Scene

Fig. 6 shows the propagated uncertainty for each MSI band due to instrument spectral characterization (input uncertainties given in Table III) for the simulated spectral desert scene.

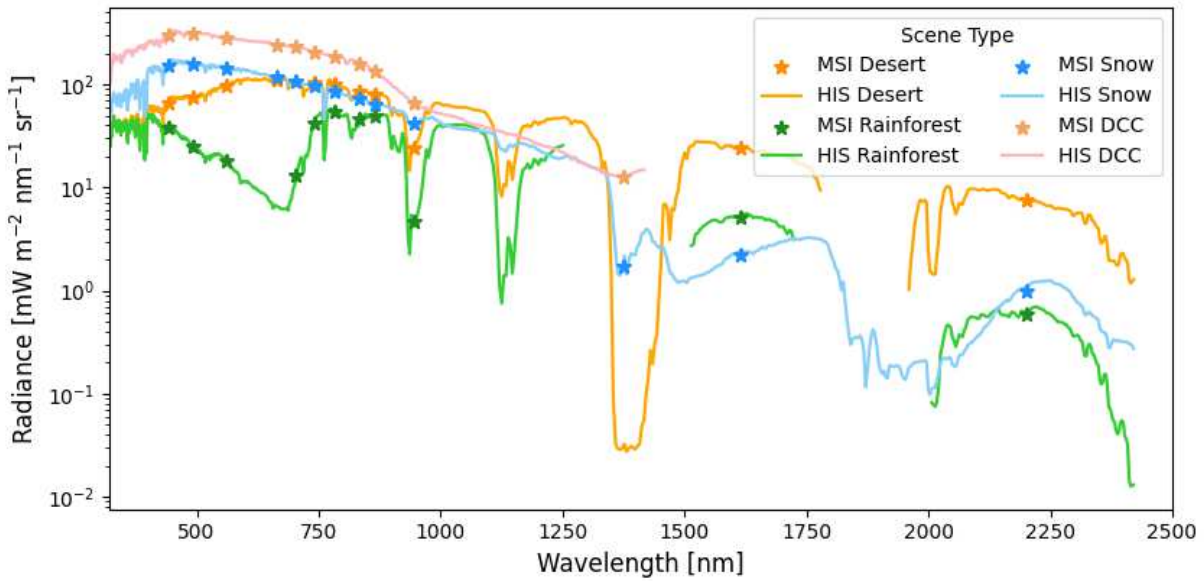


Fig. 4. Simulated TRUTHS HIS (line) and Sentinel-2A MSI (stars) TOA radiance spectra plotted on a log scale for four spectral simulated scenes: desert (orange), rainforest (green), snow (blue), DCC (peach). These spectra are TOA simulated radiances based on high-resolution data propagated to TOA and then spatially averaged over the region of interest (ROI) for visualization purposes. The unaveraged simulated scenes were used in subsequent analysis.

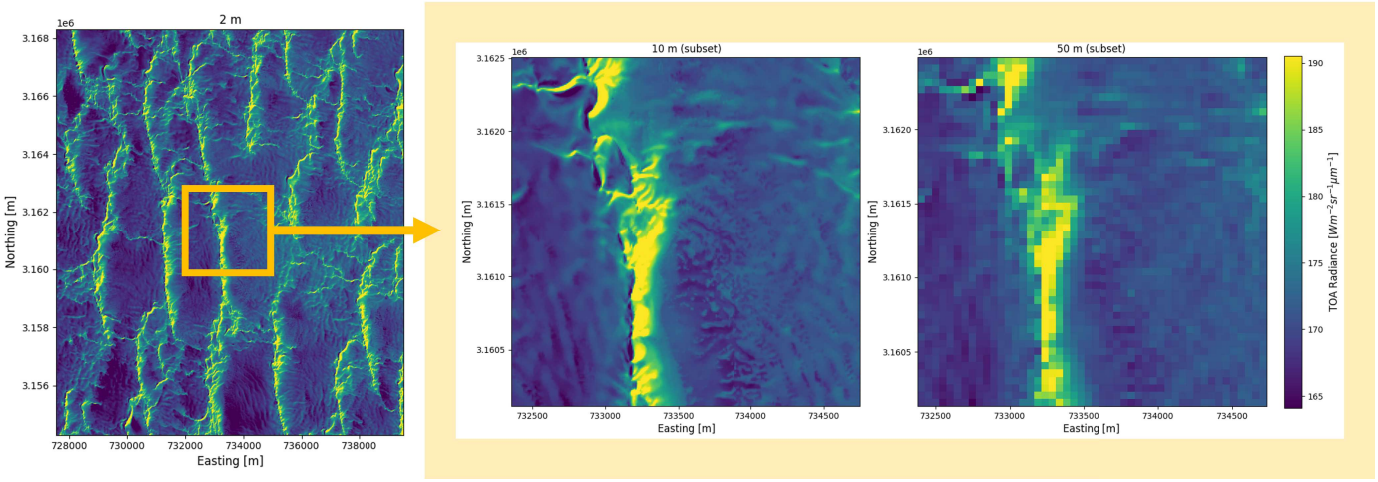


Fig. 5. 2-m resolution Airbus Pléiades TOA radiance image (CNES (2024), Distribution Airbus DS) (left) for Libya-4 desert scene for wavelength 500 nm. A subset of the image indicated by the orange square is shown for the simulated (middle) Sentinel-2 10 m and (right) TRUTHS 50-m TOA radiance images to visualize the difference in spatial resolutions of the simulated images.

The individual uncertainty contributions from systematic and random correlated errors are shown separately, and combined in quadrature to give the combined uncertainty.

Uncertainty due to the target sensor's wavelength knowledge is the dominant contributor for most bands, particularly those encompassing or bordering spectral features—such as atmospheric absorption bands (B01, B09, and B10); however, it has a limited overall impact for these applications. The smoother spectral regions—such as the visible bands B02–B04 and parts of the SWIR (e.g., B11)—exhibit lower uncertainties, generally well below 0.5%. This is in line with expectations based on the spectral smoothness of the desert reflectance spectrum (see Fig. 4). Uncertainties of this magnitude would dominate over the TRUTHS calibration uncertainty (0.3% k = 2) in their contributions to the total comparison uncertainty.

C. Simulation Uncertainty Analysis for Spectral Band Adjustment—Comparison of Different Scenes

Fig. 7 extends the analysis across multiple scene types. The results for each of the different scenes follow a similar pattern—bands encompassing smoother spectral regions show lower uncertainties, typically less than 0.5%, while those containing strong spectral features have higher uncertainties. The rainforest scene shows the greatest spectral variability, especially in the red edge region, where bands B05 and B06 show noticeably higher uncertainty than for other scenes. This reflects the steeper red edge present in vegetated surface spectra—visible in Fig. 4.

Atmospheric absorption bands (e.g., B01, B09, and B10) generally show higher uncertainty across all scenes. This is

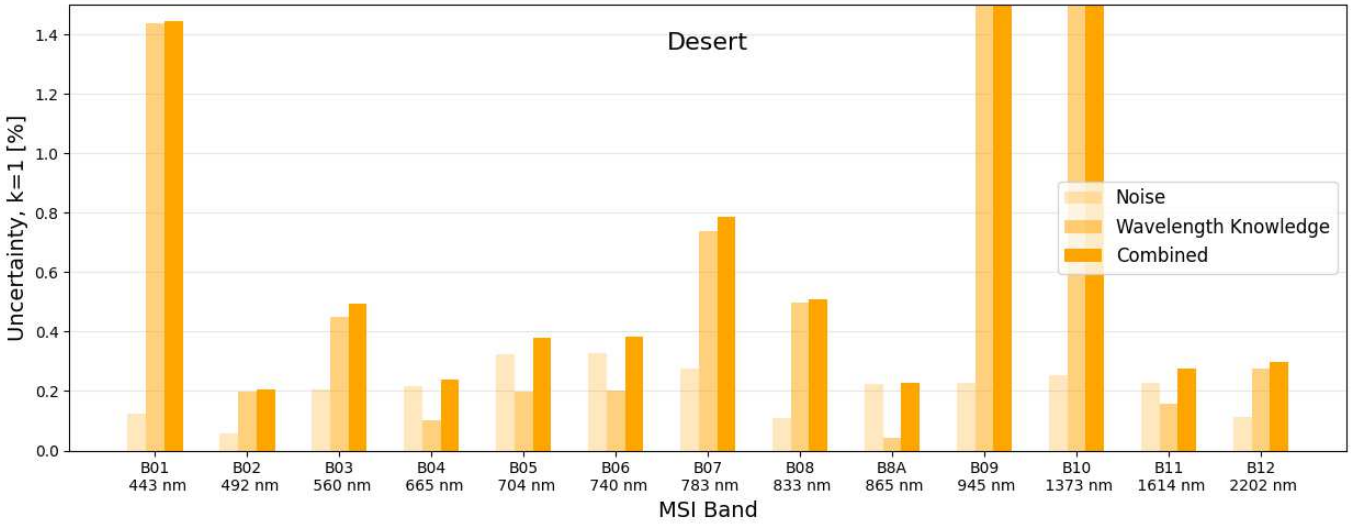


Fig. 6. Mean spectral uncertainty associated with TRUTHS HIS reconstruction of MSI bands for simulated desert scene plotted as individual uncertainty components (noise and wavelength knowledge) and combined uncertainty. The mean is calculated by averaging over all pixels in the simulated image covering an area of $\sim 5 \times 5$ km.

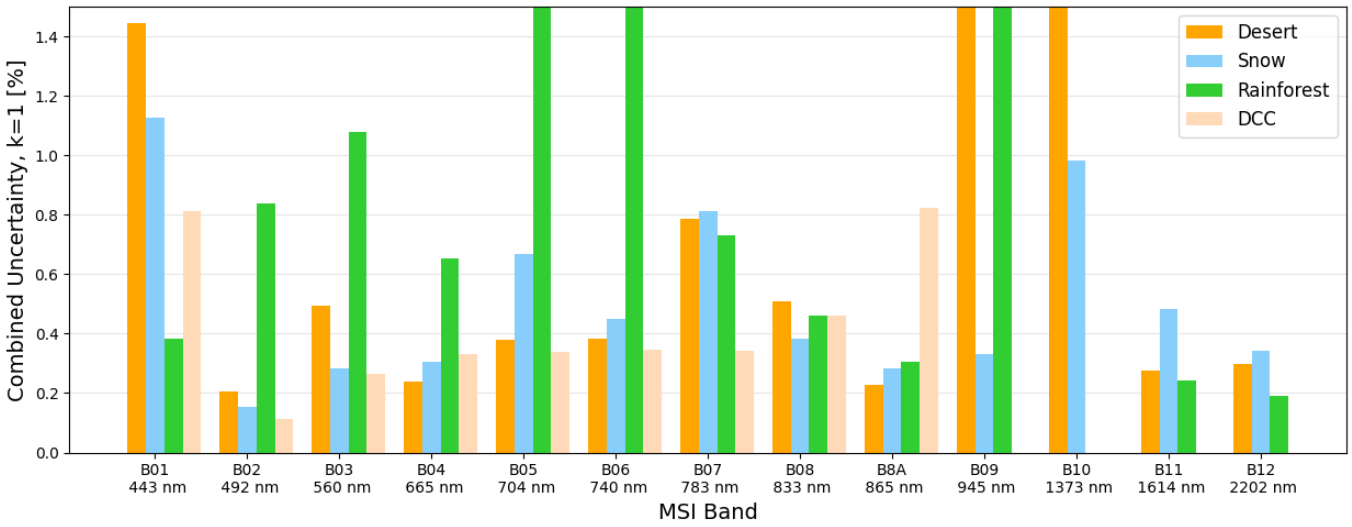


Fig. 7. Mean combined uncertainty associated with processed TRUTHS HIS radiance for each simulated scene. DCC scenes are only considered for wavelengths below 1300 nm, where they are generally regarded as spectrally flat and near-Lambertian, making them suitable targets for reflective cross-calibration. The mean is calculated by averaging over all pixels in the simulated images—each covering an area of approximately 5×5 km.

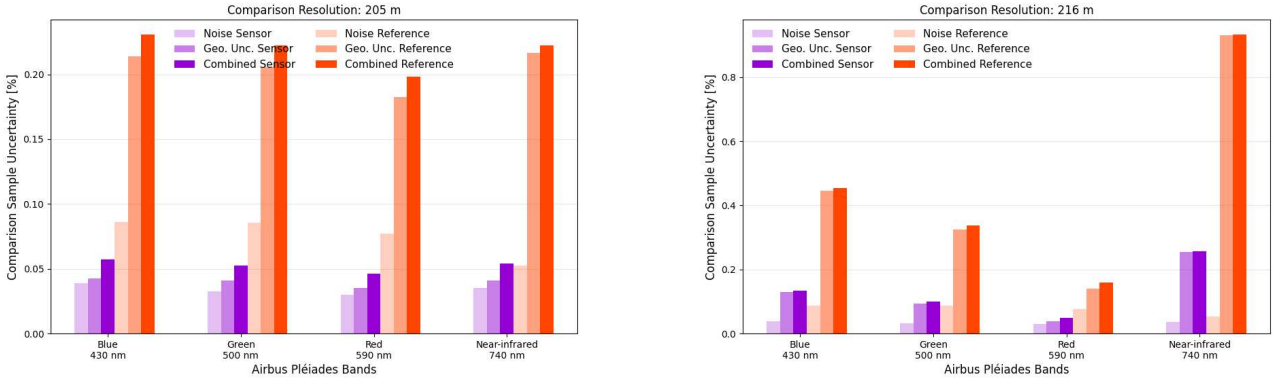
less pronounced for the DCC scene as it lies above the main atmospheric absorbers (e.g., water vapor and oxygen) in the spectral region considered (<1300 nm), so it is spectrally flatter. For all bands, there is at least one scene that results in uncertainties of less than 0.7%.

Section III-E1 will present a more detailed sensitivity analysis showing the impact of sensor wavelength knowledge on overall uncertainty, considering wavelength uncertainty values beyond those of Sentinel-2.

D. Simulation Uncertainty Analysis for Spatial Resampling

Fig. 8 shows the median uncertainty per comparison sample for each Airbus Pléiades band (assumed to be representative of typical multispectral bands like MSI) for the imagery over the Libyan desert and Amazon rainforest. The median is used to reduce the influence of outliers and provide a more robust estimate of typical uncertainty. Due to the 2-m resolution of Pléiades data, exact downscaling to 10 and 50 m was not always possible; the nearest achievable resolution was used. Consequently, the comparison grid samples are at 205 m^2 rather than 200 m^2, 20$\times$ coarser than the Sentinel 2 resolution ($k_{\mathrm{lim}} = 20$).

The uncertainty is split into components arising due to sensor noise (randomly correlated along all dimensions) and sensor geolocation uncertainty, with the combined uncertainty also shown. Within the MCM, the simulated geolocation error is added to the simulated HIS/MSI imagery by spatially shifting the Pléiades pixels before downscaling to sensor resolution, introducing a relative error between the HIS and MSI simulated images. The matchup data is then spatially resampled to the comparison grid following the algorithm



(a) (b)

Fig. 8. Median values for each component of uncertainty per comparison sample associated with the spatially resampled Sentinel-2 and TRUTHS radiance, generated using similar Pléiades spectral bands, for two scenes: (a) Libyan desert and (b) rainforest. Note that the range on the y-axis differs for the two plots.

described in Section II-C. The magnitude of the resulting component of uncertainty depends on the spatial variability within the comparison sample footprint and its surrounding area.

The rainforest scene is characterized by greater surface heterogeneity than the desert scene. To quantify this heterogeneity, the spatial homogeneity index (SHI) is used, defined as

$$\mathrm{SHI} = \frac{\sigma_{L_{\text{Pléiades, 10 m}}}}{\mu_{L_{\text{Pléiades, 10 m}}}} \tag{6}$$

where σ represents the standard deviation and μ represents the mean for the Pléiades pixels (downscaled to the finest MSI resolution, 10 m) within each comparison sample footprint. This metric is assumed to be representative of the relative spatial heterogeneity of the underlying scene within each comparison sample.

The input geolocation uncertainty is assumed to be systematic across each image, meaning all pixels are offset by the same amount. However, due to spatial variability in the underlying surface, the resulting error in the radiances of individual comparison samples is largely random between samples. As a result, this component of uncertainty diminishes as the comparison resolution becomes coarser and more spatial averaging is applied. When averaging over a 4×4 km region of the Libyan desert, the uncertainty associated with the mean radiance in the green band is <0.01% for a 50-m resolution sensor with 25m geolocation uncertainty, such as that proposed for TRUTHS at the time of writing. For a smaller 200×200 m area within the Libyan desert region, the uncertainty increases to 0.3% for samples with an SHI of 0.015—representative of the region's typical surface variability. In contrast, the rainforest scene, characterized by greater surface heterogeneity of average sample SHI of 0.08, exhibits a mean radiance uncertainty of only 0.03% over a 4×4 km area. This increases to 1.75% for a 200×200 m region.

While averaging over larger areas substantially reduces spatial uncertainties, smaller comparison resolutions (e.g., 200 m) enable more effective filtering of comparison samples for subsequent analysis. This may be beneficial in isolating homogeneous regions, that is, calibration sites, within larger heterogeneous scenes, for example.

E. Sensitivity of the Uncertainty to Sensor Characterization

The results presented in the previous section provided an estimate of the uncertainty associated with the comparison samples for TRUTHS-Sentinel-2 arising due to the effects driven by sensor characterization and specification.

In this section, the sensitivity of the comparison sample uncertainties to the target sensor wavelength knowledge and geolocation uncertainty of the sensors is investigated in more detail, extending beyond the performance characteristics of Sentinel-2. As before, this analysis is based on the simulated measurements (Figs. 4 and 5). The results focus on bands outside strong spectral absorption features (e.g., red–green–blue (RGB) and NIR) to better isolate sensor-related uncertainties. Bands affected by atmospheric absorption (notably around ~940 and 1375 nm) are excluded, as their sensitivity to scene-dependent effects and atmospheric variability makes them less suitable for cross-calibration using this methodology.

1) Sensitivity to Target Sensor Band Central Wavelength Knowledge: As highlighted in Fig. 6, the uncertainty associated with the spectral adjustment is driven by the target sensor band central wavelength knowledge. Fig. 9 isolates the impact of this effect to show the sensitivity of the output uncertainty to the wavelength knowledge for the input target sensor bands. As expected, the greater the target sensor wavelength uncertainty, the greater the uncertainty associated with the output reference sensor comparison samples. The relationship can be seen to be approximately linear for all of the nonatmospheric absorption bands for the range of wavelength uncertainties included (0.1–2.0 nm). For Sentinel-2's wavelength knowledge of 1 nm, indicated by the dashed line, the uncertainty arising due to this effect is as high as 1% for the red edge band over rainforest, and greater than 0.5% for many other bands over this scene type. One band has an effect as large as 0.8% over the desert, but the rest, while substantial, are <0.5%.

In contrast, for the wavelength knowledge of TRUTHS HIS of 0.05 nm ($k = 1$), indicated by the dotted line, the uncertainty

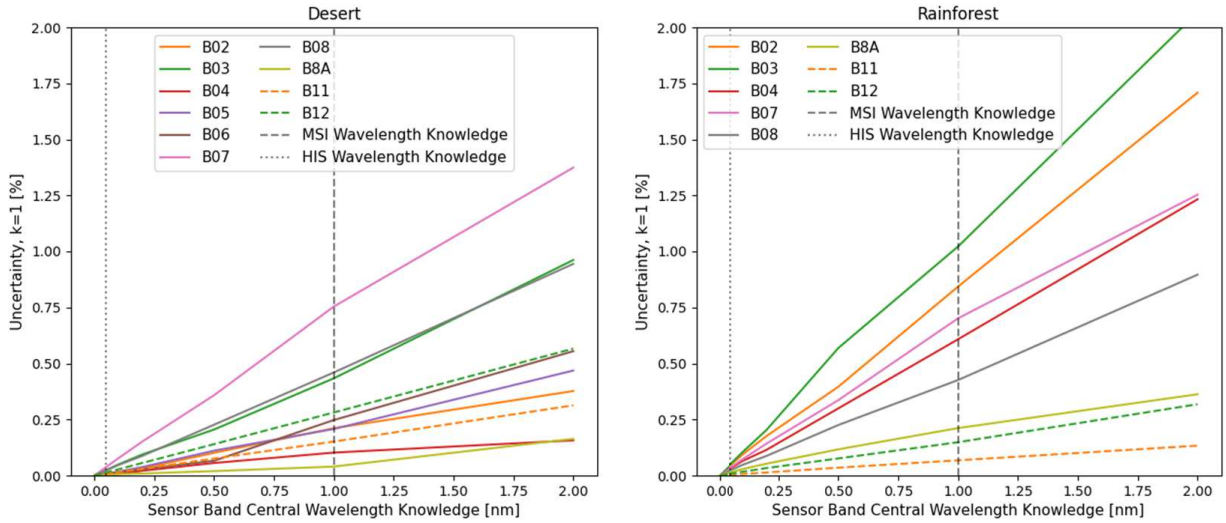


Fig. 9. Sensitivity of uncertainty associated with spectrally adjusted TRUTHS HIS imagery ($k = 1$) to target sensor wavelength knowledge for desert (left) and rainforest (right). Vertical dotted and dashed lines represent the wavelength knowledge for HIS bands (nominally 0.05 nm at the time of writing) and MSI bands (nominally 1 nm) [34], respectively. Only nonatmospheric absorption bands are plotted for each scene, as these are less affected by scene-specific atmospheric variability and provide a clearer assessment of sensor-related uncertainties.

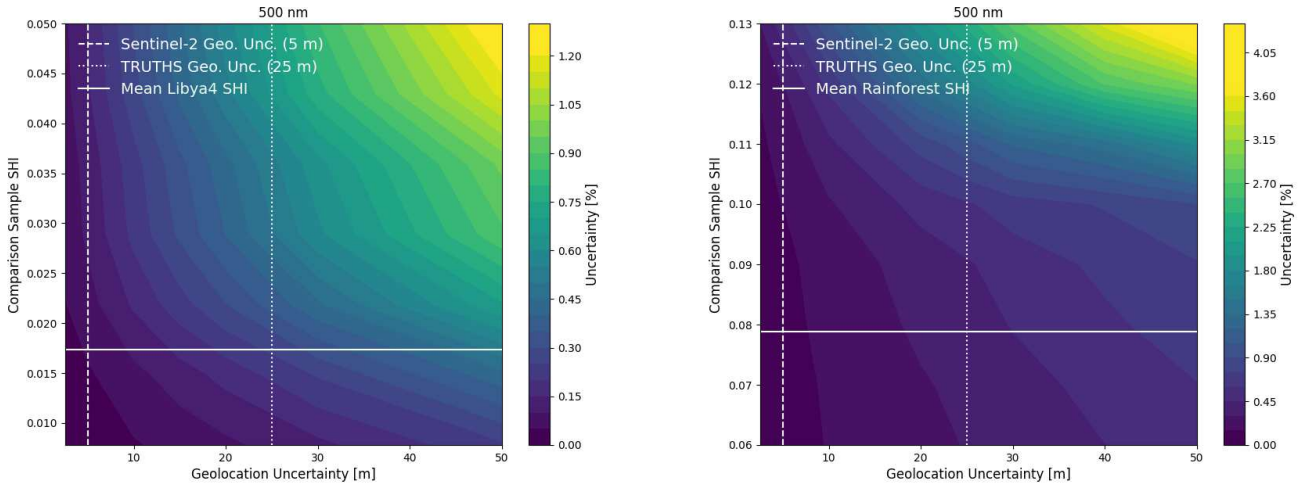


Fig. 10. Propagated uncertainty ($k = 1$) associated with the comparison samples as a function of sensor geolocation uncertainty and SHI of the underlying surface at 205-m comparison resolution for the 500-nm band. Vertical dotted/dashed lines represent the nominal target geolocation uncertainty for TRUTHS (25 m) and Sentinel-2 (5 m) for Earth-view radiance measurements. Horizontal lines indicate the mean SHI for the ROI considered: (left) Libya-4 and the surrounding desert and (right) rainforest scene in the Amazon.

is <0.05%. For bands where spectral features are driven by the surface rather than the atmosphere, this uncertainty may be reduced by aggregating matchups over different scene types with different underlying spectral shapes.

2) Sensitivity to Geolocation Uncertainty: Fig. 10 shows the uncertainty associated with the spatially processed comparison samples as a function of sensor geolocation uncertainty and SHI of the comparison sample footprint for the Airbus Pléiades green (500 nm) band. The SHI is calculated as (6).

The relationship between geolocation uncertainty and the resulting uncertainty in the comparison samples is approximately linear within the range of geolocation uncertainties considered for all MSI bands (based on results for additional bands not shown here). Scenes with higher underlying spatial heterogeneity exhibit not only greater absolute uncertainty, but also an increased sensitivity to input geolocation error.

The average SHI of comparison samples in the simulated Libyan desert image is ~0.02. For sensor geolocation uncertainty of 5 and 25 m, this corresponds to an uncertainty of <0.05% and <0.25%, respectively, for a single comparison sample area (205 × 205 m) for the RGB (approximately 460–680 nm) and NIR (approximately 700 880 nm).

F. Total Comparison Uncertainty for Sensor Characterization Effects

The spectral and spatial uncertainties are assumed to be uncorrelated for the error effects considered in this study. Following the GUM guidelines [35], the total comparison uncertainty is calculated by summing these components in quadrature. Since the noise component of the spectral uncertainty is already propagated using MCM to derive the random

TABLE V

COMBINED UNCERTAINTY (k = 1) VALUES FOR THE SIMULATED SENTINEL-2 OBSERVATIONS OVER THE LIBYAN DESERT SCENE FOR THE RGB BANDS. RESULTS ARE SHOWN FOR THE 205-M-RESOLUTION CASE, AND A 4 × 4 KM REGION OF INTEREST (ROI). ALL UNCERTAINTIES ARE EXPRESSED AS PERCENTAGES, WITH u_{TOTAL} REPRESENTING THE QUADRATURE SUM OF THE CONTRIBUTING SPECTRAL AND GEOLOCATION TERMS ONLY

Spectral Channel	Estimated Radiance $Wm^{-2}sr^{-1}m^{-1}$	$u_{\text{wav, spectral}}$ [%]	$u_{\text{geo, spatial}}$ [%]	$u_{\text{noise, spatial}}$ [%]	u_{total} [%]
205 m×205 m ROI					
B02 (Blue)	152	0.21	0.21	0.09	**0.31**
B03 (Green)	195	0.44	0.21	0.09	**0.50**
B04 (Red)	220	0.11	0.18	0.08	**0.23**
4 km×4 km ROI					
B02 (Blue)	152	0.21	0.01	0.00	**0.21**
B03 (Green)	195	0.44	0.01	0.00	**0.44**
B04 (Red)	220	0.11	0.01	0.00	**0.11**

TABLE VI

COMBINED UNCERTAINTY COMPONENTS (k = 1) FOR RGB SENTINEL-2 BANDS ACROSS DIFFERENT SCENE TYPES FOR AN INDIVIDUAL COMPARISON SAMPLE OF ~ 200 × 200 M. ALL UNCERTAINTIES ARE EXPRESSED AS PERCENTAGES, WITH u_{TOTAL} REPRESENTING THE QUADRATURE SUM OF THE CONTRIBUTING SPECTRAL AND GEOLOCATION TERMS ONLY

Spectral Channel	Desert			Snow			Rainforest		
	u_{wav}	u_{geo}	$\mathbf{u}_{\text{total}}$	u_{wav}	u_{geo}	$\mathbf{u}_{\text{total}}$	u_{wav}	u_{geo}	$\mathbf{u}_{\text{total}}$
B02 (Blue)	0.21	0.23	**0.31**	0.17	0.09	**0.19**	0.86	0.46	**0.98**
B03 (Green)	0.44	0.22	**0.50**	0.29	0.09	**0.30**	1.09	0.34	**1.14**
B04 (Red)	0.11	0.20	**0.23**	0.30	0.08	**0.31**	0.64	0.16	**0.66**

component of the spatial uncertainty (see Section II-D), it is not included separately.

The total uncertainty is, therefore, calculated as

$$u_{\text{total}} = \sqrt{u^2_{\text{wav, spectral}} + u^2_{\text{geo, spatial}} + u^2_{\text{noise, spatial}}}. \quad (7)$$

Table V gives the uncertainty values for the simulated Libyan desert scenes for the RGB bands for an individual comparison sample of ~ 200 × 200 m, and, below, for an ROI of ~ 4 × 4 km. The RGB bands are presented here as they lie outside major atmospheric absorption features and, being observed at MSI's finest spatial resolution of 10 m, are also the most sensitive to spatial effects, providing a representative view of sensor-related uncertainties. The correlation form (systematic, random) of errors influences their relative importance over different spatial and temporal scales [38]. Randomly correlated errors, for example, errors arising due to geolocation uncertainty, cancel out when considering many matchups in a comparison or using a coarser comparison resolution, whereas systematic errors, such as wavelength error within a scene, become more dominant.

The results indicate that both spectral and spatial effects contribute significantly to the total comparison uncertainty at the level of individual comparison samples. However, when samples are aggregated over a larger area or number of matchup opportunities, spatial uncertainties are substantially reduced, leaving systematic spectral uncertainties as the dominant contributor.

Table VI presents the spectral, spatial, and combined (k = 1) uncertainty values for individual comparison samples across all three scene types: desert (Libya-4), rainforest (Amazon), and Antarctic snow (near Dome-C), again for the RGB bands.

The rainforest has the highest total uncertainty—up to 1.14% in the green band for individual samples. In contrast, the snow scene, the brightest and most homogeneous surface, has total uncertainties of only 0.3% for these bands. These total uncertainties are largely driven by the spectral component of uncertainty, even for an individual sample over a small area. When averaging over a larger 4 × 4 km region, spatial uncertainties are reduced for all scenes and can be considered negligible.

IV. DISCUSSION

This work has examined how the design of a SITSat—including accurate wavelength knowledge, low radiometric uncertainty, and hyperspectral bands—can enhance the performance of cross-calibration for multispectral sensors in the solar reflective domain. These benefits are exemplified by the upcoming TRUTHS mission. The resulting improvements are shown to be subject to the characteristics and characterization of the SITSat and the sensor under test. Using a detailed simulation study, the impact of sensor characterization on cross-calibration uncertainty has been quantified, with a focus on spectral and spatial effects. For typical desert scenes, the combined uncertainty associated with the radiance at the comparison sample level was found to be around 0.2%–0.5% for the RGB bands. Over larger 4 × 4 km areas, the spatial contribution is reduced and can be considered negligible to the overall uncertainty budget. For rainforest scenes, which have greater surface heterogeneity,

uncertainties are higher—up to 1.2% at 200-m resolution for the RGB bands—again dominated by spectral effects, with spatial uncertainty rapidly diminishing when samples are aggregated for comparison as for the desert scene. The spectral component arising due to the wavelength knowledge of the target sensor is shown to dominate, a result consistent with previous studies [12]. However, the magnitude of this component of uncertainty is significantly reduced to <0.5% when using the TRUTHS hyperspectral sensor as the reference over desert scenes, compared to values of up to 2.5% for multispectral reference sensors used in previous work. These results demonstrate that, while the effects considered here are nonnegligible, their combined contributions are commensurate with the expected uncertainty of the TRUTHS HIS reference sensor (0.3% $k = 2$), and significantly lower than current cross-calibration capabilities (typically 3%–5%). With robust—and ideally enhanced—characterization of the target sensors, these findings reinforce the potential of TRUTHS-enabled cross-calibration to deliver a step change in cross-calibration performance.

In the adjustment performed to align the observations spectrally, the systematic uncertainty in this study is driven by wavelength knowledge of the sensor under test, Sentinel-2 MSI. These results extend previous studies on the impact of characterization for intercalibration with a multispectral reference sensor [12] by exploring the advantages a hyperspectral reference sensor like the TRUTHS HIS can provide. The findings highlight the importance of accurate spectral characterization of the target sensor to fully leverage the enhanced cross-calibration performance that SITSats like TRUTHS can offer. As shown in Fig. 9, the comparison uncertainty has an approximately linear relationship with the target sensor's central wavelength uncertainty. Bands encompassing significant spectral absorption features are most sensitive to these spectral uncertainties, as expected.

These results illustrate the need for careful scene selection, taking the target wavelength range into particular consideration to minimize the impact of systematic spectral uncertainties. For example, while the spectral component of comparison uncertainty may be just over a few percent in the red edge over vegetated scenes, it is considerably lower (<0.4%) over desert and snow scenes. Utilizing matchups across a diverse set of scenes can further reduce spectral uncertainty for bands containing spectral features that vary between scene types. This is because, although errors arising from sensor wavelength knowledge are assumed to be systematic within a given scene, Fig. 7 illustrates that the uncertainty can vary across different scenes based on the underlying spectral characteristics. These characteristics are influenced by scene-specific spectral features, such as the red edge in vegetated areas, and atmospheric absorption features that impact multiple scene types. The correlation of wavelength uncertainty across global matchups—and its impact on comparison uncertainty—as well as the potential to detect and characterize wavelength shifts, will be explored in future work.

In the spatial resampling simulation analysis, the uncertainty is driven by the geolocation uncertainty of both sensors, with values up to 0.25% and 0.5% for the RGB bands at a comparison resolution of ~200 m for the Libyan desert and rainforest scenes, respectively. Fig. 10 shows that the resulting comparison uncertainty is highly dependent on the heterogeneity of the underlying scene, evaluated at the smallest scale available using the simulated Sentinel-2-resolution pixels (10 m). The findings demonstrate that for comparisons at the spatial scales of a few HIS pixels, matchups—or regions of pixels within a matchup—should be filtered by surface homogeneity to reduce the impact of geolocation errors. Although averaging over larger areas is shown to reduce spatial uncertainties, this approach is not always the most effective or feasible. Smaller comparison resolutions, for example, 200 m as in this study, allow for more targeted filtering, which is essential in situations where surface heterogeneity or observational effects like geometry, clouds, or timing mismatches are present. This can help the selection of homogeneous regions maximizing the utility of data in all images, but particularly those considered optimum for calibration (i.e., homogeneous, filtered for large geometry/timing mismatches). By not averaging radiances over large areas initially, the ability to filter out less suitable data is preserved, improving the accuracy of subsequent analyses and cross-calibration.

This study considers only true nadir, simultaneous matchups, thereby neglecting the impact of geometry and temporal mismatches. These effects are expected to be partially mitigated when considering a globally diverse set of matchups under real-world conditions. Previous studies show that defining thresholds for geometry and temporal mismatches can help in mitigating the impact of these effects [13]. More broadly, many sources of comparison uncertainty, such as these observational mismatches and the spatial-related effects considered in this work, are expected to give rise to errors that are randomly correlated across different matchups. Moreover, TRUTHS' polar orbit provides a unique opportunity to maximize the diversity of matchup events across various times, latitudes, and viewing angles, allowing for the aggregation of comparison data from a wide range of scenarios. Assuming stability of sensor calibration over time, this can significantly reduce uncertainties arising from temporal and angular differences, as there will not be a systematic bias between satellites due to the observational geometry, as is the case for matchups between satellites with sun-synchronous orbits.

This work has focused on the impact of sensor characterization on the overall comparison uncertainty by considering a limited range of scenarios. Future work will explore a wider range of contributing error effects, such as BRDF, and so on, to enable a more comprehensive uncertainty budget for matchup comparisons employing data from a SITSat reference such as TRUTHS.

V. Conclusion

The TRUTHS climate satellite mission is designed to act as a "gold-standard" reference for EO sensors, facilitating an upgrade to the wider EO system. The mission has been designed to optimize the performance of cross-calibration, for example, through the orbit selection and band specification.

The unprecedented low uncertainty that will be offered by TRUTHS, and other SITSats, will ensure that

cross-comparison uncertainty is no longer be dominated by the measurement uncertainty of the reference observations. This work evaluates the impact of sensor design and characterization effects on comparison uncertainty, aiming to ensure that the resulting uncertainty is commensurate with the expected uncertainty of the TRUTHS reference sensor (0.3% $k = 2$) and remains generally lower than typical cross-calibration uncertainties (3%–5%). At this stage, only spectral and spatial effects are considered, assuming ideal SNO conditions. Future work will evaluate the degree to which mismatches in viewing and illumination geometries between sensors can be tolerated while still meeting the calibration requirements for a particular application or set of applications. For example, an uncorrected viewing/illumination zenith angle mismatch of <5° results in an error of only <1% in the TOA radiance over desert calibration sites [13].

Accurate preflight characterization and in-flight monitoring of target sensor wavelength knowledge are required for optimal performance of cross-calibration with a reference sensor like TRUTHS HIS across all scenes. Notably, bands encompassing significant spectral features (e.g., absorption bands and red-edge) are particularly sensitive to the uncertainties considered due to the underlying spectra within these bands. Cross-calibration of target sensors using direct comparison with a reference sensor, as demonstrated here, may therefore not provide as much benefit for bands encompassing spectral features (e.g., atmospheric absorption features). Careful scene selection can partially mitigate these effects for some bands.

In summary, this work highlights the necessity of accurate sensor characterization—particularly wavelength knowledge—to fully realize the benefits of space-based SIT-Sat missions like TRUTHS. While the methodology here focused on idealized, simultaneous-nadir matchups, future work will broaden this analysis to more globally diverse conditions, accounting for effects such as viewing geometry and polarization. Together, these developments will establish a comprehensive uncertainty budget for cross-calibration and support more accurate, traceable EO from space.

ACKNOWLEDGMENT

The authors would like to thank Dave Doelling, NASA Langley Research Center, for providing SCIAMACHY data used in the simulation of the DCC scene. They declare no conflict of interest.

REFERENCES

[1] V. Masson-Delmotte, "Climate change 2021: The physical science basis," in *Proc. Contribution Work. Group I 6th Assessment Rep. Intergovernmental Panel Climate Change*, vol. 2, no. 1, Geneva, Switzerland, 2021, p. 2391.

[2] N. Fox et al., "Accurate radiometry from space: An essential tool for climate studies," *Phil. Trans. Roy. Soc. A, Math., Phys. Eng. Sci.*, vol. 369, no. 1953, pp. 4028–4063, Oct. 2011.

[3] H. Boesch et al., "Si-traceable spacebased climate observation system: A CEOS and GSICS workshop, national physical laboratory, London, U.K.," Nat. Phys. Lab., London, U.K., Tech. Rep., 2022, pp. 9–11.

[4] N. Fox et al., "Traceable radiometry underpinning terrestrial-and helio-studies (truths)," *Adv. Space Res.*, vol. 32, no. 11, pp. 2253–2261, 2003.

[5] Y. Shea et al., "Clarreo pathfinder: Mission overview and current status," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Waikoloa, HI, USA, Sep. 2020, pp. 3286–3289.

[6] P. Zhang et al., "Development of the Chinese space-based radiometric benchmark mission LIBRA," *Remote Sens.*, vol. 12, no. 14, p. 2179, Jul. 2020.

[7] J. E. Martin, N. P. Fox, and P. J. Key, "A cryogenic radiometer for absolute radiometric measurements," *Metrologia*, vol. 21, no. 3, pp. 147–155, Jan. 1985.

[8] S. Saunier et al., "Sen2Like: Paving the way towards harmonization and fusion of optical data," *Remote Sens.*, vol. 14, no. 16, p. 3855, Aug. 2022.

[9] G. Chander, T. J. Hewison, N. Fox, X. Wu, X. Xiong, and W. J. Blackwell, "Overview of intercalibration of satellite instruments," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 3, pp. 1056–1080, Mar. 2013.

[10] H. Xu et al., "Radiometric cross-calibration of sentinel-2B MSI with HY-1C SCS based on the near simultaneous imaging of common ground targets," *Opt. Exp.*, vol. 31, no. 3, pp. 3619–3637, 2023.

[11] M. M. Farhad, M. Kaewmanee, L. Leigh, and D. Helder, "Radiometric cross calibration and validation using 4 angle BRDF model between Landsat 8 and Sentinel 2A," *Remote Sens.*, vol. 12, no. 5, p. 806, Mar. 2020.

[12] G. Chander, D. L. Helder, D. Aaron, N. Mishra, and A. K. Shrestha, "Assessment of spectral, misregistration, and spatial uncertainties inherent in the cross-calibration study," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 3, pp. 1282–1296, Mar. 2013.

[13] J. Gorroño, M. P. Solé, N. Fox, L. Guanter, T. August, and T. Fehr, "Global assessment of directional effects in the inter-calibration of optical satellite instruments with the truths mission," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5536313.

[14] C. Cao et al., "Establishing the Antarctic dome C community reference standard site towards consistent measurements from Earth observation satellites," *Can. J. Remote Sens.*, vol. 36, no. 5, pp. 498–513, Oct. 2010.

[15] W. Wu et al., "The angular correction algorithm for the intercalibration of satellite instruments using CLARREO pathfinder as a reference," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5507211.

[16] J. Gorroño, A. C. Banks, N. P. Fox, and C. Underwood, "Radiometric inter-sensor cross-calibration uncertainty using a traceable high accuracy reference hyperspectral imager," *ISPRS J. Photogramm. Remote Sens.*, vol. 130, pp. 393–417, Aug. 2017.

[17] C. Lukashin, B. A. Wielicki, D. F. Young, K. Thome, Z. Jin, and W. Sun, "Uncertainty estimates for imager reference inter-calibration with CLARREO reflected solar spectrometer," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 3, pp. 1425–1436, Mar. 2013.

[18] A. Wu, X. Xiong, Z. Jin, C. Lukashin, B. N. Wenny, and J. J. Butler, "Sensitivity of intercalibration uncertainty of the CLARREO reflected solar spectrometer features," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 9, pp. 4741–4751, Sep. 2015.

[19] C. Cao, X. Wu, A. Wu, and X. Xiong, "Improving the SNO calibration accuracy for the reflective solar bands of AVHRR and MODIS," *Proc. SPIE*, vol. 6684, Aug. 2007, Art. no. 668408.

[20] P. Henry, G. Chander, B. Fougnie, C. Thomas, and X. Xiong, "Assessment of spectral band impact on intercalibration over desert sites using simulation based on EO-1 hyperion data," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 3, pp. 1297–1308, Mar. 2013.

[21] K. Shimaoka, M. Kinoshita, K. Fujii, and T. Tosaka, "Evaluation of measurement data-supplement 1 to the guide to the expression of uncertainty in measurement-propagation of distributions using a Monte Carlo method," *JCGM*, vol. 101, pp. 1–82, 2008.

[22] H. Cosnefroy, M. Leroy, and X. Briottet, "Selection and characterization of Saharan and Arabian desert sites for the calibration of optical satellite sensors," *Remote Sens. Environ.*, vol. 58, no. 1, pp. 101–114, Oct. 1996.

[23] P. Teillet, J. A. Barsi, G. Chander, and K. J. Thome, "Prime candidate Earth targets for the post-launch radiometric calibration of space-based optical imaging instruments," *Proc. SPIE*, vol. 6677, pp. 304–315, Sep. 2007.

[24] N. Mishra, D. Helder, A. Angal, J. Choi, and X. Xiong, "Absolute calibration of optical satellite sensors using Libya 4 pseudo invariant calibration site," *Remote Sens.*, vol. 6, no. 2, pp. 1327–1346, Feb. 2014.

[25] B. Fougnie and R. Bach, "Monitoring of radiometric sensitivity changes of space sensors using deep convective clouds: Operational application to PARASOL," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 3, pp. 851–861, Mar. 2009.

[26] N. Lamquin, V. Bruniquel, and F. Gascon, "Sentinel-2 L1C radiometric validation using deep convective clouds observations," *Eur. J. Remote Sens.*, vol. 51, no. 1, pp. 11–27, Jan. 2018.

[27] D. R. Doelling, "Personal communication," NASA Langley Res. Center, Hampton, VA, USA, Tech. Rep., 2023.

[28] S. Cogliati et al., "The PRISMA imaging spectroscopy mission: Overview and first performance analysis," *Remote Sens. Environ.*, vol. 262, Sep. 2021, Art. no. 112499.
[29] C. Emde et al., "The libradtran software package for radiative transfer calculations (version 2.0. 1)," *Geosci. Model Develop.*, vol. 9, no. 5, pp. 1647–1672, 2016.
[30] N. Mishra, M. Haque, L. Leigh, D. Aaron, D. Helder, and B. Markham, "Radiometric cross calibration of Landsat 8 operational land imager (OLI) and Landsat 7 Enhanced Thematic Mapper Plus (ETM+)," *Remote Sens.*, vol. 6, no. 12, pp. 12619–12638, Dec. 2014.
[31] L. V. Rizzo et al., "Long term measurements of aerosol optical properties at a primary forest site in Amazonia," *Atmos. Chem. Phys.*, vol. 13, no. 5, pp. 2391–2413, Mar. 2013.
[32] C. Tomasi et al., "Characterization of the atmospheric temperature and moisture conditions above dome c (Antarctica) during austral summer and fall months," *J. Geophys. Res., Atmos.*, vol. 111, no. D20, Oct. 2006, Art. no. D20305.
[33] B. N. Holben et al., "AERONET—A federated instrument network and data archive for aerosol characterization," *Remote Sens. Environ.*, vol. 66, no. 1, pp. 1–16, Oct. 1998.
[34] P. Martimort, V. Fernandez, V. Kirschner, C. Isola, and A. Meygret, "Sentinel-2 MultiSpectral imager (MSI) and calibration/validation," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2012, pp. 6999–7002.
[35] Joint Committee for Guides in Metrology, "Evaluation of measurement data—Guide to the expression of uncertainty in measurement," vol. 100, pp. 1–116, 2008.
[36] P. De Vis and S. E. Hunt, "Comet toolkit (community metrology toolkit)," Nat. Phys. Lab., Teddington, U.K., Tech. Rep., 2025.
[37] P. Sajadi, M. Gholamnia, S. Bonafoni, and F. Pilla, "Spectral-based regression model for destriping of abnormal pixel values in PRISMA hyperspectral image," *Eur. J. Remote Sens.*, vol. 55, no. 1, pp. 622–643, Dec. 2022.
[38] C. J. Merchant et al., "Uncertainty information in climate data records from Earth observation," *Earth Syst. Sci. Data*, vol. 9, no. 2, pp. 511–527, Jul. 2017.



Madeline Stedman received the M.Sci. degree in natural sciences from the University of Exeter, Exeter, U.K., in 2020. She is currently pursuing the joint Ph.D. degree with the National Physical Laboratory (NPL), Teddington, U.K., and Imperial College London, London, U.K.

Since 2020, she has been a Scientist at the Climate and Earth Observation Group, NPL. She has focused on sensor-to-sensor cross-calibration for Earth observation satellites, particularly in the visible and near-infrared spectral domains. In particular, she contributes to the development of the European Space Agency's Traceable Radiometry Underpinning Terrestrial- and Helio-Studies (TRUTHS) mission, where she is involved in algorithm development and performance modeling to support in-flight calibration of Earth observing sensors.



Samuel E. Hunt received the M.Phys. degree in physics from Durham University, Durham, U.K., in 2014.

Since 2016, he has been a Senior Scientist at NPL, Teddington, U.K., specializing in satellite Earth Observation (EO) metrology, with a focus on improving the quality and trust in EO data to support climate action. He has led efforts to apply metrological techniques to the generation of high-quality fundamental data records (FDRs), particularly from visible and thermal radiometric imagery. His work spans sensor modeling, uncertainty analysis, intersatellite calibration, and data quality assurance. He has been involved in the development of the ESA TRUTHS mission, where he contributes to algorithm development, covering radiometric processing, performance modeling, and cross-calibration.



Pieter De Vis received the joint Ph.D. degree in galaxies from Ghent University, Ghent, Belgium, and the University of Canterbury, Christchurch, New Zealand, in 2016, with a focus on cosmic dust.

He is a Senior Scientist in the Climate and Earth Observation Group, NPL, Teddington, U.K. His focus is on uncertainty propagation, calibration, and validation of satellite reflectance products, and providing metrologically robust reference data from in situ reference networks. He leads the development of Python tools for uncertainty propagation, atmospheric correction, and the application of vicarious calibration. Before joining NPL in 2019, he worked in Astronomy, a topic that shares many concepts with Earth observation.



Richard Bantges received the Ph.D. degree in atmospheric physics from Imperial College London, London, U.K., in 2000, focus on the cirrus cloud radiative properties in the thermal infrared.

He is the Scientific Project Manager and a National Center for Earth Observation Research Scientist with the Department of Physics, Imperial College London. He is currently a member of the Mission Advisory Group for ESA's Earthwatch mission, TRUTHS, providing expert advice on the requirements of observations for climate applications and a climate benchmark. He has accrued more than 25 years' experience studying the Earth's energy balance, principally exploiting satellite observations in conjunction with radiative transfer modeling. He has made important contributions to many Earth observation and space missions for both the European Space Agency (ESA) and the National Aeronautics and Space Administration (NASA).



Helen Brindley is a Professor of Earth observation with the Department of Physics, Imperial College London, London, U.K. Her primary research involves measuring and understanding variability in the Earth's radiation budget. Her team develops ground-based, airborne, and satellite instrumentation in support of this work. She is a PI of the Geostationary Earth Radiation Budget (GERB) Instrument, flying on the Meteosat Second Generation of Satellites, and leads U.K. activities related to ESA's Far-infrared Outgoing Radiation Understanding and Monitoring (FORUM) Earth Explorer mission as part of the Mission Advisory Group. She is a Divisional Director with the UK's National Center for Earth Observation, with responsibility for coordinating research into the Earth's energy and water cycles. She has a particular interest in exploring how the high-resolution temporal, spatial, and spectral behavior of Earth's outgoing radiation links to processes driving climate variability and change.



Nigel Fox received the B.Sc. degree in astronomy and physics from the University College London, London, U.K., in 1981, and the Ph.D. degree from the University of London, London, in 1997.

He is currently an NPL Fellow and the Chief Scientist of Earth Observation Climate and Optical Radiometry, NPL, Teddington, U.K., and a Visiting Professor with the Department of Meteorology, University of Reading, Reading, U.K. At University College London, he developed new primary optical radiation measurement scales and techniques that led to an improvement of nearly two orders of magnitude in the accuracy of many radiometric measurements. Over the last three decades, he has focused on satellites, leading to innovation in preflight and post-launch calibration and validation techniques, culminating as the Lead Scientist of the ESA TRUTHS satellite mission.

Dr. Fox is the Chair of the Infrared, Visible, and Optical Sensors (IVOS) Subgroup of CEOS WGCV.